Chapter 4 | Blanchard, Macroeconomics 9e

Session 1: Money Demand + Interest Rate Determination

Session 2: Banks, Reserves & Liquidity Traps

73-103 Principles of Macroeconomics | CMU-Q

Carnegie Mellon University Qatar

Financial Markets I

Why Money & Interest Rates Matter

Why Should You Care About Financial Markets?

The real-world impact on your daily life

- **Interest rates affect everything: mortgages, car loans, business investment**
- **Financial markets channel savings into productive investment**
- **Central bank decisions ripple through the entire economy**
- **Financial markets: bonds, stocks, currencies, mutual funds**

Example: Global Bond Market
~$130 Trillion
Larger than all stock markets

KEY INSIGHT Financial markets are the transmission mechanism between central bank policy and the real economy.

SECTION 4-1

The Demand for Money

Why do people hold money instead of interest-bearing assets?

- The money vs. bonds tradeoff
- Determinants of money demand
- The money demand function

Money vs. Bonds: The Fundamental Tradeoff

The simplifying assumption: only two financial assets

MONEY		BONDS
✓ **Yes: spend it now (liquid)**	**Use it for transactions?**	✗ No: must sell it first
✗ No: 0% return	**Pays interest?**	✓ **Yes: earns the interest rate *i***
✓ **None: value fixed in dollars**	**Risk of price change?**	✗ Yes: its price can change
Cash, checking accounts *Qatar: QAR cash, QNB checking*	**Examples**	T-bills, government bonds *Qatar: sovereign bonds*

Example: $10,000 for one year, *i* = 5%

As money: $10,000, and you can spend it any time

As a bond: $10,500 next year (+ $500 interest)

Opportunity cost of holding money = *i* × $10,000 = $500

LIQUIDITY convenience ⟷ **RETURN** interest

Higher *i* → convenience costs more → people hold less money (next slide).

KEY INSIGHT The choice between money and bonds depends on the tradeoff between liquidity (convenience) and return (interest).

What Drives Money Demand?

Two key determinants

1. Level of Transactions (Nominal Income, $Y)

- More income = more spending = need more money
- Money demand increases proportionally with $Y
- Think: Holiday shopping season vs. January

↑ $Y → ↑ Md

2. Interest Rate on Bonds (i)

- Higher i = higher opportunity cost of holding money
- People shift wealth from money to bonds
- Money demand decreases as i rises

↑ i → ↓ Md

The Money Demand Function

Formalizing the relationship

$$Md = \$Y \times L(i)$$

Money Demand = Nominal Income × Liquidity Function

Md
Money Demand
(how much money people want to hold)

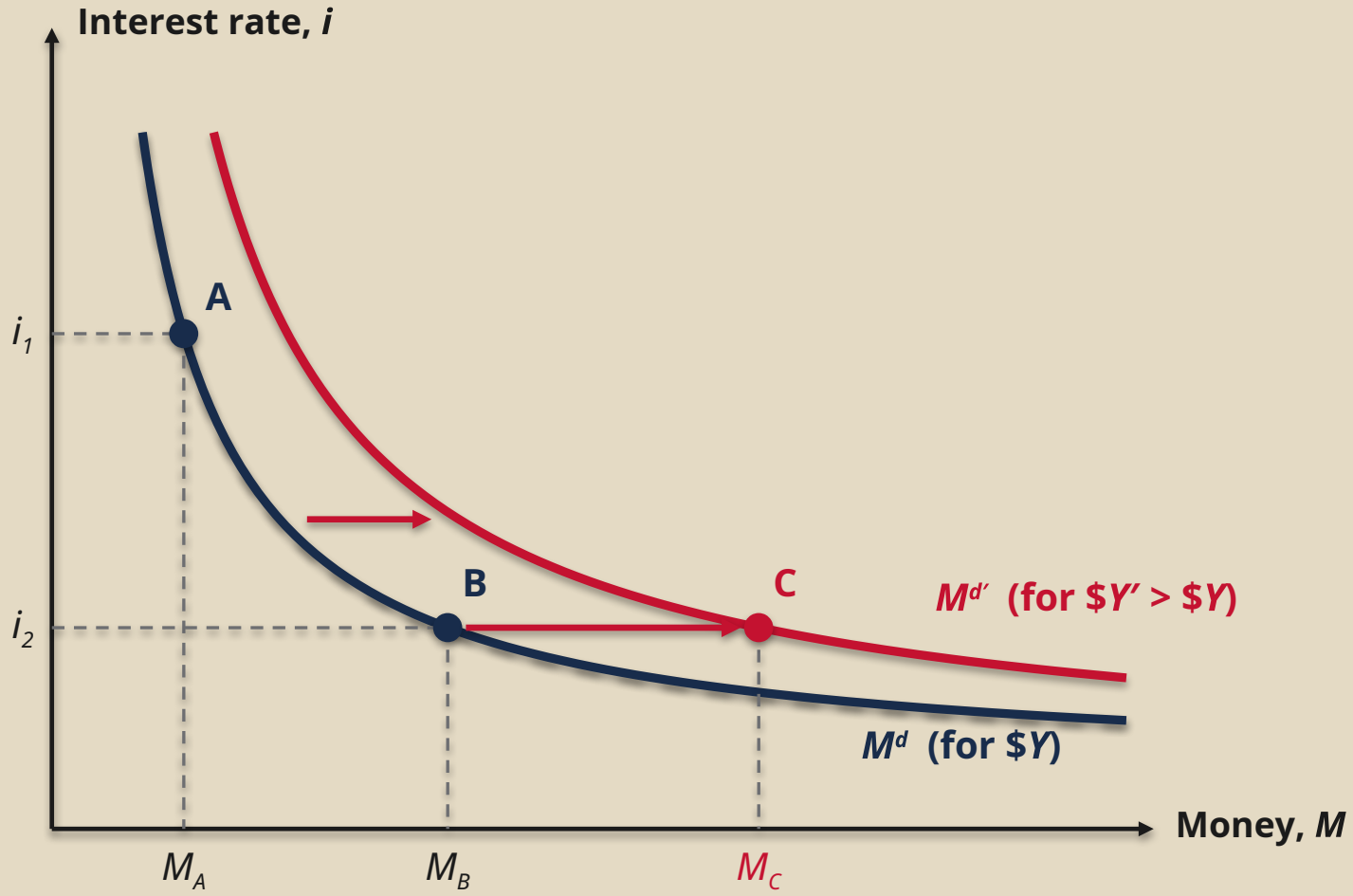
$Y
Nominal Income
(proportional: more income = more demand)

L(i)
Liquidity Function
(decreasing in i: higher i = less demand)

KEY INSIGHT L(i) is a decreasing function: as interest rates rise, people hold less money and more bonds.

Figure 4-1: The Demand for Money

Money demand as a function of the interest rate



Downward slope: A → B

Higher *i* = higher opportunity cost of holding money. As *i* falls, people hold more money.

Scenario: income rises ($Y ↑)

More income = more transactions = more money demanded at every interest rate.

Higher income shifts M^d right: at every interest rate, people want to hold more money.

Semantic Traps: Money ≠ Income ≠ Wealth

Getting the vocabulary right

Illustration: one household

Income
$5,000 per month

FLOW · measured per period

Saving
$1,000 per month

FLOW · per month

Spending
$4,000 per month

WEALTH = $200,000

MONEY = $3,000
cash + checking

Everything you own, measured today

Other assets = $197,000
bonds, stocks, house, …

STOCK · measured on a date

Units test

A flow has a time unit (per month, per year). A stock is measured on a date (e.g., 31 December).

Trap 1

"She makes a lot of money." → She has a high INCOME (a flow).

Trap 2

"He has a lot of money." → He has a lot of WEALTH (a stock).

In this course

Money = cash + checking deposits: the assets you can use for transactions right away.

KEY INSIGHT

Money is a STOCK (what you hold now); income is a FLOW. Money demand = how much of your wealth you choose to hold as money.

Think-Pair-Share: What Happens to Money Demand?

ACTIVITY

1. Qatar's nominal GDP doubles due to a new LNG expansion. What happens to Md?

2. The QCB raises interest rates from 5% to 7%. What happens to Md?

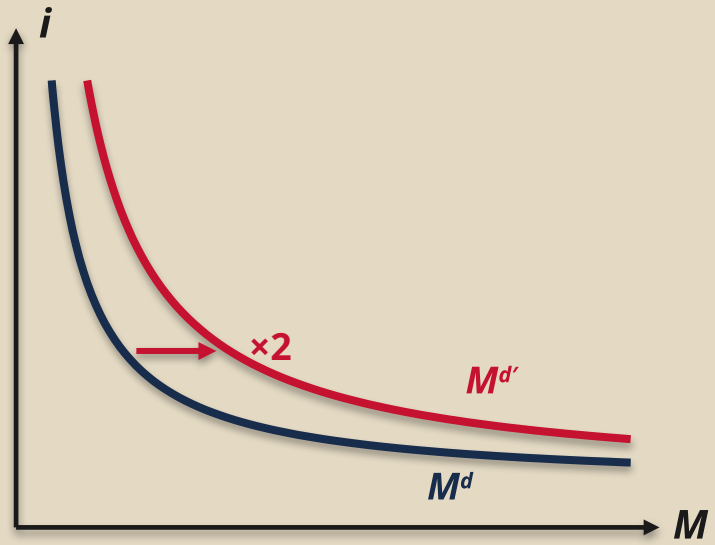
3. A new mobile payment app makes it easier to convert bonds to cash instantly. What happens to L(i)?

Discuss in pairs (3 minutes). Be ready to explain using the Md = $Y × L(i) framework.

Think-Pair-Share: Answers

Shift the curve, or move along it?

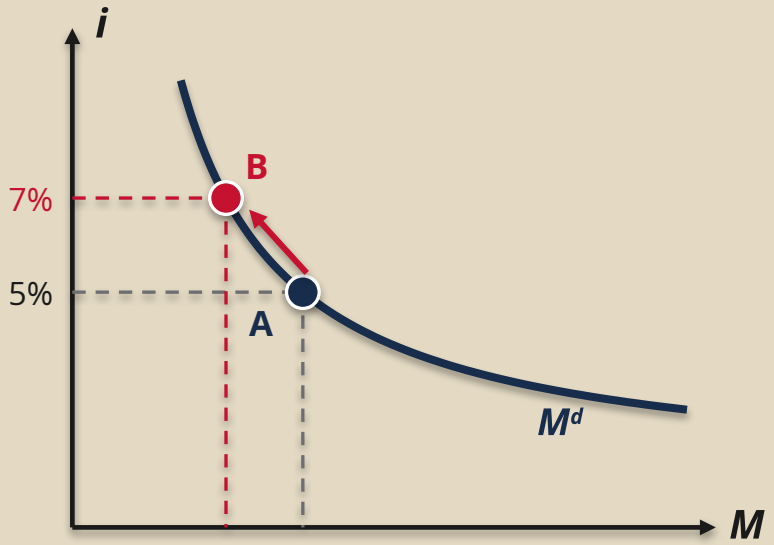
Q1 · Nominal GDP doubles (LNG expansion)



M^d shifts RIGHT

$M^d = \$Y \times L(i)$: if $\$Y$ doubles, money demand doubles at every interest rate.

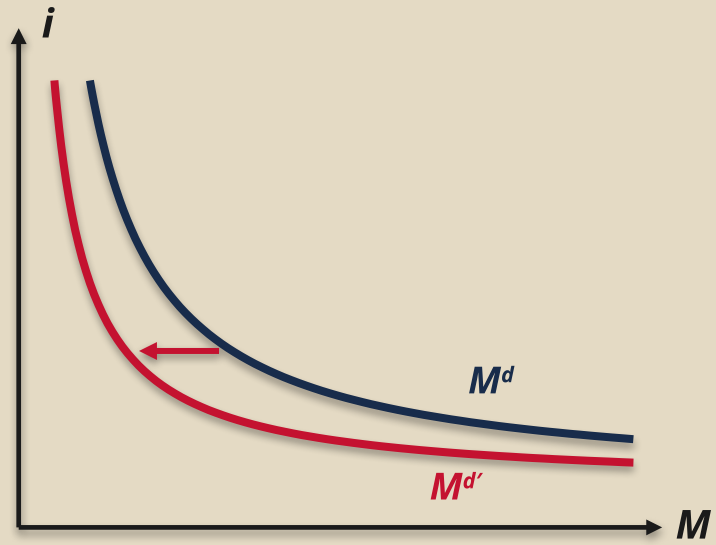
Q2 · QCB raises *i* from 5% to 7%



Movement ALONG M^d (A → B)

Higher *i* = higher opportunity cost of holding money, so people hold less. The curve itself does not move.

Q3 · Payment app: bonds → cash instantly



M^d shifts LEFT: $L(i)$ falls

If bonds turn into cash instantly, people need to hold less money at every interest rate.

$Y changes, or a new payment technology → the M^d curve SHIFTS. *i* changes → we MOVE ALONG it.

Quick Check: Money Demand

Test your understanding

True or False: Money demand rises with income.

?

True or False: Higher interest rates increase money demand.

?

If $\$Y = \$1{,}000$ and $L(i) = 0.5$, what is M^d?

?

Commit to an answer for each one before we reveal.

Quick Check: Money Demand

Test your understanding

True or False: Money demand rises with income.	**TRUE**	More income → more transactions → more money needed. M^d is proportional to $\$Y$.
True or False: Higher interest rates increase money demand.	**FALSE**	Bonds pay i, money pays 0. Higher i = higher opportunity cost, so people hold less money: $L(i)$ falls as i rises.
If $\$Y$ = \$1,000 and $L(i)$ = 0.5, what is M^d?	**\$500**	M^d = \$1,000 × 0.5 = \$500. $L(i)$ = 0.5 means people hold money worth half of their nominal income.

Follow-up 1: i rises so that $L(i) = 0.4$ → M^d = \$1,000 × 0.4 = \$400 (a movement along M^d)

Follow-up 2: $\$Y$ doubles to \$2,000 → M^d = \$2,000 × 0.5 = \$1,000 (the M^d curve shifts right)

KEY INSIGHT Money demand: proportional to \$Y, decreasing in i. The formula Md = \$Y × L(i) captures both relationships.

From Demand → Equilibrium

We know what determines money demand.
Now: How does the interest rate get determined?

Answer: Where money supply meets money demand.

SECTION 4-2

Determining the Interest Rate I

The simplest model: money supply, money demand, and equilibrium

- Money market equilibrium
- Effects of income and money supply changes
- Open market operations

Equilibrium in the Money Market

Where supply meets demand

$$Ms = Md \quad \rightarrow \quad Ms = \$Y \times L(i)$$

Money Supply (set by central bank) = Money Demand

Money Supply (Ms)

Set by the central bank as a policy decision. Vertical line on graph.

Money Demand (Md)

Determined by income ($Y) and interest rate (i). Downward-sloping curve.

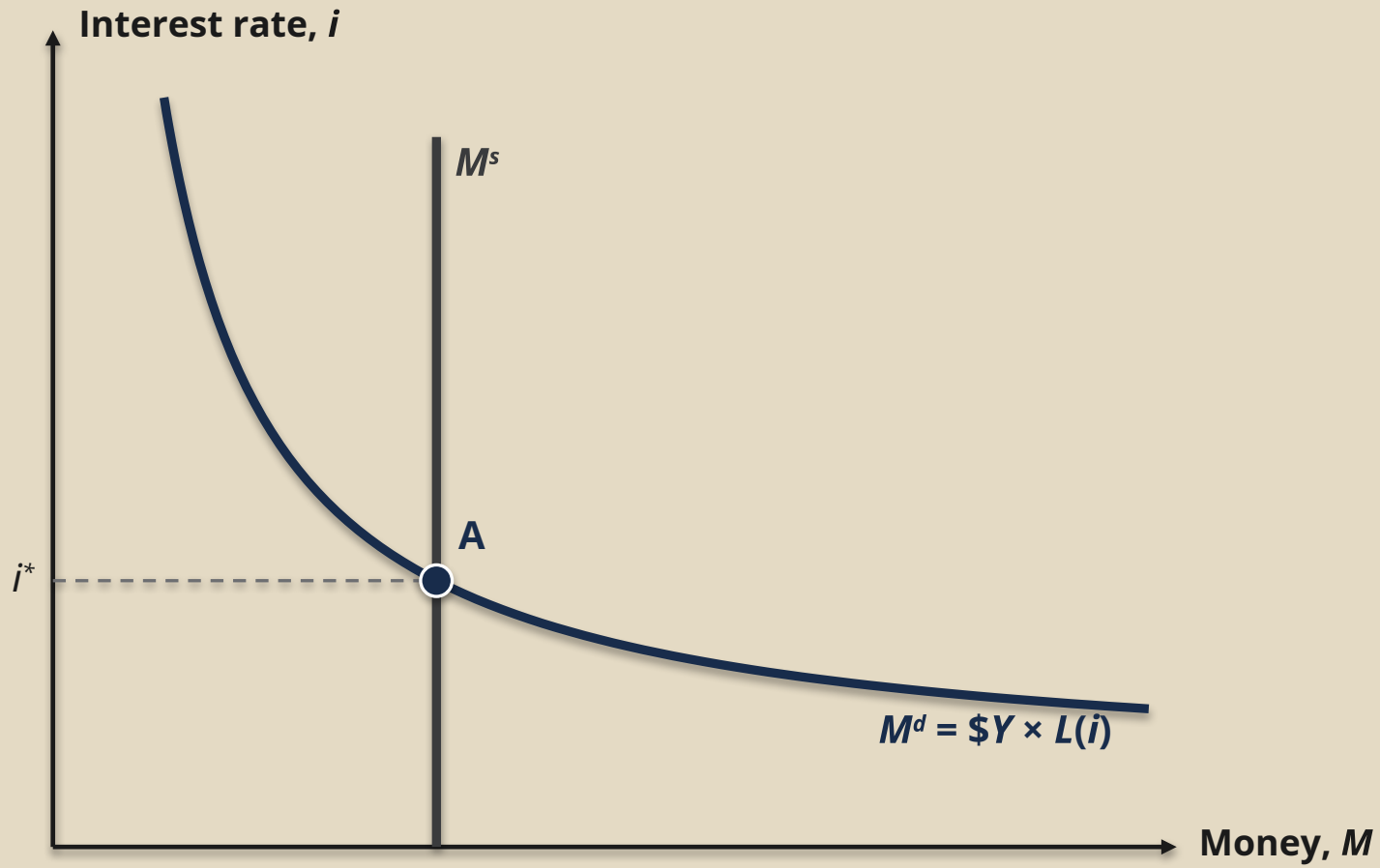
Equilibrium (i*)

The interest rate where Ms = Md. Found at intersection.

KEY INSIGHT The central bank controls Ms. Given $Y, the equilibrium interest rate i* adjusts so people willingly hold exactly Ms.

Figure 4-2: Determination of the Interest Rate

Money market equilibrium



M^d slopes down

Higher i = higher opportunity cost, so people hold less money.

M^s is vertical

The central bank fixes the money supply (a policy decision). It does not depend on i.

Equilibrium A: $M^s = M^d$

The intersection pins down the interest rate i^*.

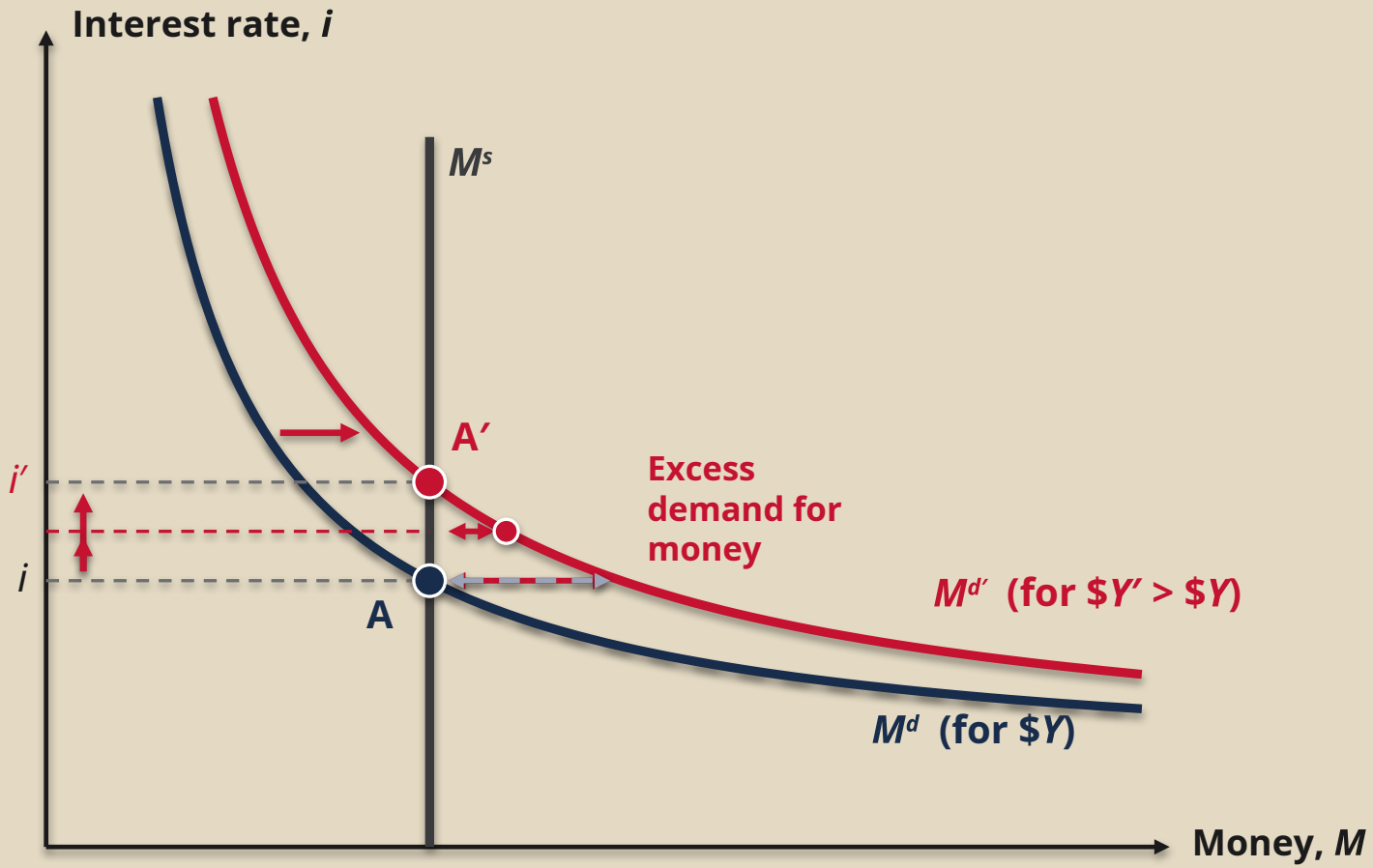
The central bank controls M^s. Given $\$Y$, i^* adjusts so people willingly hold exactly M^s.

Figure 4-3: What Happens When Income Rises?

Effect of an increase in nominal income on the interest rate

Figure 4-3: What Happens When Income Rises?

Effect of an increase in nominal income on the interest rate



Shock: income rises ($*Y* ↑)

More transactions → more money demanded at every *i*.

Adjustment through the bond market

Excess demand for money → SELL bonds

$$i = \frac{\$100 - P_B}{P_B}$$

P_B ↓

⇒ *i* ↑

P_B = price of the bond today
\$100 = what the bond pays in one year
i = the bond's yield = our interest rate

e.g. P_B falls \$96 → \$95, so *i* rises 4.2% → 5.3%

Higher *i* → holding money costs more → quantity demanded falls along $M^{d'}$.

Selling stops at A′: $M^{d'} = M^s$ at the higher i'. People willingly hold the existing money.

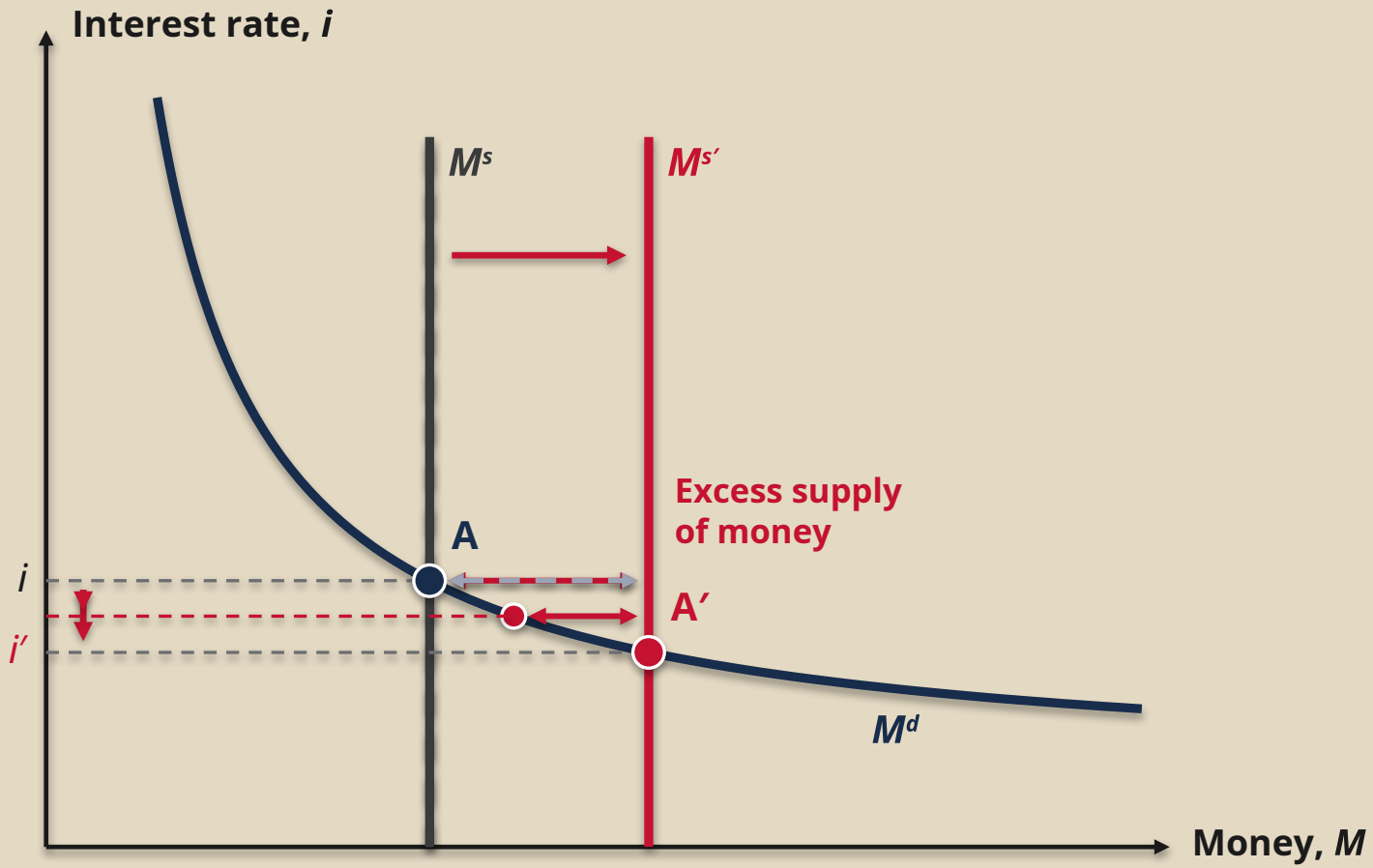
Real world: an economic boom pushes interest rates up, unless the central bank expands M^s.

Figure 4-4: What Happens When Money Supply Rises?

Effect of an increase in money supply on the interest rate

Figure 4-4: What Happens When Money Supply Rises?

Effect of an increase in money supply on the interest rate



Policy: central bank raises M^s

Money supply shifts right; money demand is unchanged.

Adjustment through the bond market

Excess supply of money → BUY bonds

$$i = \frac{\$100 - P_B}{P_B}$$

P_B ↑ ⇒ i ↓

P_B = price of the bond today
\$100 = what the bond pays in one year
i = the bond's yield = our interest rate

e.g. P_B rises \$95 → \$96, so i falls 5.3% → 4.2%

Lower i → holding money costs less → quantity demanded rises along M^d.

Buying stops at A′: $M^d = M^{s'}$ at the lower i'. People willingly hold the new money.

This is how central banks stimulate the economy!